



Incident handler's journal

Incident Overview: Ransomware Attack on Healthcare Provider

Incident Summary:

This report documents a high-impact security incident involving a small U.S.-based primary-care health clinic. On a Tuesday morning at approximately 9:00 a.m., the organization experienced a total disruption of business operations when employees were locked out of critical systems.

Technical Root Cause:

Initial forensics indicate the breach was triggered by a phishing campaign. An employee inadvertently downloaded a malicious attachment, which executed a ransomware payload across the local network.

Impact Assessment:

The attack resulted in the full encryption of patient medical records and the delivery of a ransom note from an organized threat group demanding payment for a decryption key. This incident forced a temporary cessation of patient care services, highlighting a critical need for enhanced detection and response capabilities.

The following Incident Handler's Journal provides a structured breakdown of the event for remediation and audit purposes.

Date: January 2, 2026	Entry: #1
Description	Initial analysis of ransomware attack targeting a small health care clinic resulting in critical data encryption and business disruption
Tool(s) used	Incident handler journal for documenting and technical reporting.
The 5 W's	• Who: an organised group of unethical hackers targeting the healthcare industry. • What: A ransomware attack initiated via a targeted phishing email with a malicious attachment • When: Tuesday morning at approximately 9:00 a.m. • Where A small U.S based primary-care health clinic. • Why : To extort money from the clinic in exchange for a decryption key to restore access to patient records.
Additional notes	The incident highlights a critical failure in email security and employee training. Future remediation must include phishing simulation and off-site backup verification..

Methodology and Documentation Process

The following table is my Incident Handler's Journal, a core documentation tool used throughout the incident response lifecycle. I derived these entries through a combination of:

- **Scenario-Based Analysis:** Investigating the business and patient-care impact of a ransomware attack on a healthcare clinic.
- **Technical Lab Practicals:** Using a virtual Linux environment (Qwiklabs) to perform live packet sniffing with tcpdump.
- **Evidence Gathering:** Isolating source/destination IP addresses and verifying the **TCP three-way handshake** to ensure network protocol integrity.

This journal serves as a verifiable log of my ability to detect, analyze, and document security threats following the NIST Incident Response Framework.

Date: January 3, 2026.	Entry: #2
Description	Conduct a live packet capture to analyse IP headers, port numbers, and TCP flags.
Tool(s) used	Tcpdump And Linux Bash Terminal.
The 5 W's	● Who: Security Analyst. ● What: Captured and filtered network packets to view IP headers and TCP flags. ● When: During the "Sound the Alarm" technical lab. ● Where: Hosted Qwiklabs virtual machine environment. ● Why: To identify indicators of compromise (IoC) and verify standard network communication protocols.
Additional notes	Successfully isolated source and destination IP addresses and confirmed the TCP three-way handshake (SYN, SYN-ACK, ACK).

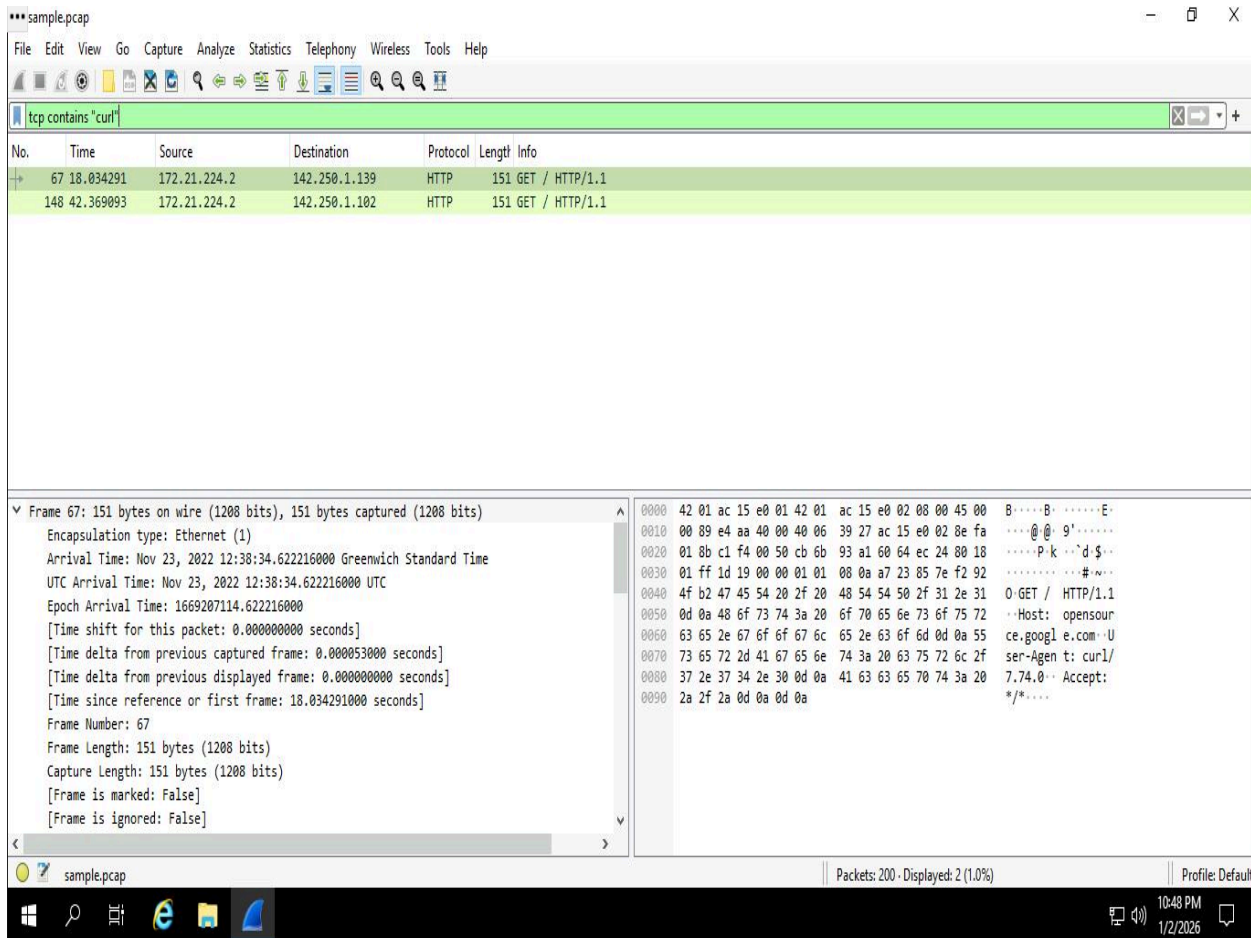


Figure 1: Applying display filters in Wireshark to isolate specific HTTP traffic generated by the 'curl' command.

By applying specific protocol filters, I reduced the complexity of the capture to focus on unauthorized or suspicious 'curl' commands. This step was critical for identifying the specific data payloads being sent from the source IP 172.21.224.2 to external destinations

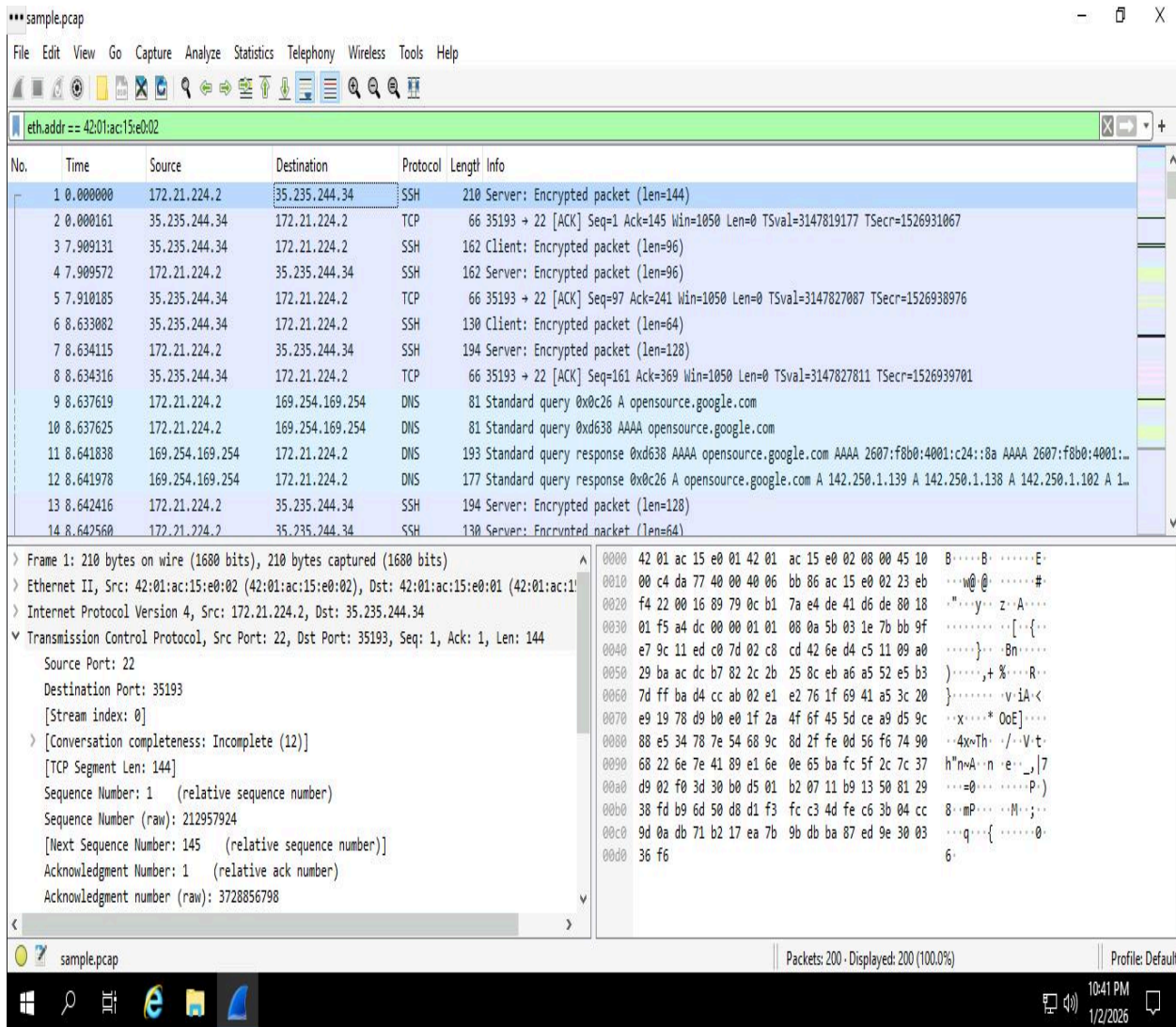


Figure 2: Comprehensive traffic log showing multi-protocol communication and DNS resolution.

I analyzed the packet sequence to confirm that DNS queries successfully resolved before the establishment of TCP sessions. The presence of encrypted SSH packets indicates secure administrative traffic was present, which I compared against the baseline of authorized network behavior.

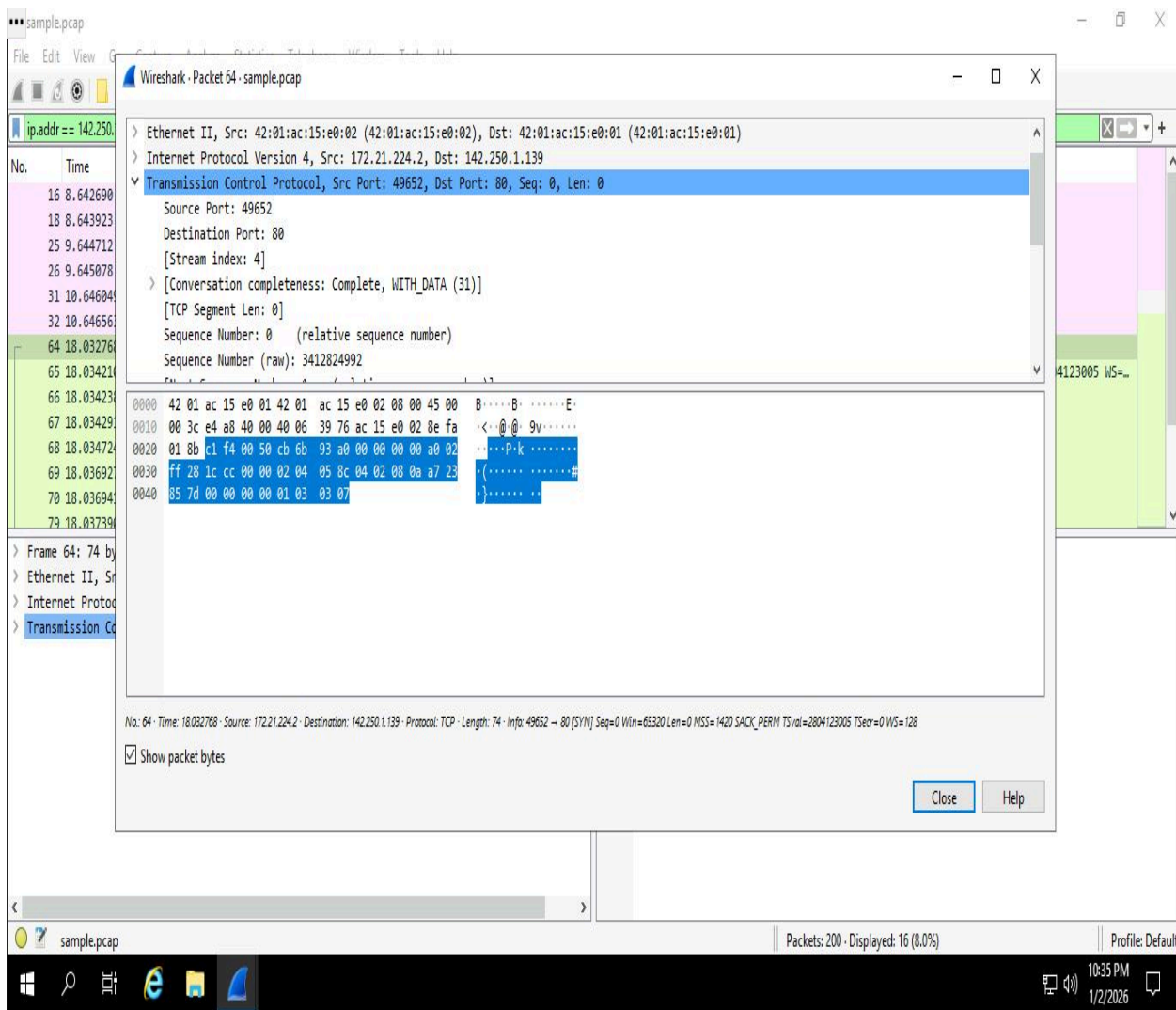


Figure 3: Technical inspection of the TCP header identifying source/destination ports and conversation completeness.

Deep packet inspection (DPI) of the TCP layer revealed a connection on Port 80, indicating unencrypted web traffic. By examining the sequence and acknowledgement numbers, I verified that the connection was 'Complete, WITH_DATA,' confirming that a full handshake and data exchange occurred

Network Traffic Analysis: Packet Inspection via tcpdump

Project Overview

As a network analyst operating within a Linux virtual machine environment, I performed a live traffic interception and forensic analysis task. The primary objective was to demonstrate technical proficiency in command-line interface (CLI) networking tools by identifying active hardware interfaces, capturing real-time packet data, and performing deep-packet inspection (DPI) to verify protocol integrity.

Technical Workflow

The investigation followed a systematic four-task methodology:

Interface Discovery: Utilized `ifconfig` and `tcpdump -D` to map available network pathways.

Live Observation: Executed verbose captures on the `eth0` interface to monitor active communication strings.

Traffic Generation: Used `curl` to produce specific HTTP (Port 80) traffic, ensuring a controlled data set for capture.

Forensic Filtering: Applied advanced command-line switches (`-nn`, `-r`, `-X`) to analyze packet headers and payloads in hexadecimal and ASCII formats.

Technical Competencies & Skill Mapping

By completing this lab, I have demonstrated mastery in the following core cybersecurity domains:

Network Security Monitoring: Proficiency in capturing live packet streams to establish a baseline of normal network activity.

Command Line Interface (CLI): Advanced capability in managing security tasks within a Linux environment without the use of a GUI.

Deep Packet Inspection (DPI): Ability to analyze raw data payloads in hexadecimal and ASCII to detect anomalies or hidden threats.

Forensic Tooling: Practical experience with `tcpdump` and `ifconfig` for data collection and network troubleshooting.

Date: January 3, 2026	Entry: #3
Description	Conducted network interface identification and utilized tcpdump to capture and filter live HTTP traffic for forensic analysis.
Tool(s) used	Ifconfig, tcpdump, curl.
The 5 W's	• Who: Network Analyst (analyst user). • What: Captured 5 live packets from eth0 and filtered Port 80 traffic into a .pcap file. • When: During the "Sound the Alarm" detection and response lab. • Where: Local Linux Virtual Machine. • Why: To verify network communication integrity and inspect packet payloads for anomalies.
Additional notes	Successfully utilized the -X flag to inspect packet data in hexadecimal and ASCII formats. This confirmed the ability to perform deep-packet inspection (DPI) in a CLI environment.

Incident Response Investigation: Phishing Alert A-2703

Project Overview

As a level-one Security Operations Center (SOC) analyst for a financial services firm, I conducted a formal investigation into a high-risk phishing alert (Ticket A-2703). The following report documents the execution of the organization's Phishing Incident Response Playbook v1.0 to identify indicators of compromise (IoCs), evaluate threat severity, and perform necessary remediation and escalation steps.

Exhibit A: Malicious Email Body (Alert A-2703)

From: Def Communications <76tguyhh6tgftrt7tg.su>

Subject: Re: Infrastructure Egnieer role

Body: "Hello, I am writing to you regarding the infrastructure egnieer role. I have attached my application and the password for the file is paradise10789."

Malicious File Hash: 54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

Date: January 3, 2026.	Entry: #4
Description	Managed a phishing incident response for Alert A-2703 involving a malicious file hash download ("bfsvc.exe") by an employee targeted via social engineering.
Tool(s) used	Phishing Playbook v1.0, Phishing Flowchart, SHA256 Hash Verification.
The 5 W's	• .Who: An employee at Inergy targeted by an external sender ("Def Communications"). • What: The user received a job-application phishing email and downloaded a password-protected malicious attachment. • When: Wednesday, July 20, 2022, at 09:30:14 AM. • Where: Corporate mail server and employee endpoint. • Why: To deliver a malicious payload while bypassing standard security filters through encrypted archives.
Additional notes	Artifact analysis of the email in Exhibit A revealed multiple red flags, including a mismatched sender domain and the misspelled subject line 'Egnieer'. After confirming the 'bfsvc.exe' attachment hash as malicious, I followed Phishing Playbook Step 3.2 to escalate the ticket for further remediation.

Technical Competencies Demonstrated

Incident Lifecycle Management: Proficiency in moving an incident through identification, investigation, and escalation phases.

Compliance & Governance: Strict adherence to organizational playbooks and flowcharts to maintain security standards.

Threat Identification: Capability to identify social engineering red flags (typos, sender mismatches) and verify malicious artifacts like SHA256 hashes.

Lab 5: Intrusion Detection Analysis with Suricata

Objective:

Investigate network security events by examining Suricata intrusion detection logs and interpreting rule-based triggers within a Linux-based environment.

Date: January 3, 2026.	Entry: #5 Examining Suricata Alerts and Logs
Description	Conducted a forensic review of network traffic by running Suricata against a sample.pcap file. Analyzed both standard alerts in fast.log and detailed JSON telemetry in eve.json to verify signature triggers.
Tool(s) used	Suricata IDS, Linux CLI, cat, ls, jq
The 5 W's	• Who: Cybersecurity Analyst / SOC Team • What: Identified "GET on wire" HTTP alerts and parsed log metadata. • When: During active network traffic inspection using read-pcap mode. • Where: Communication between local IP 172.21.224.2 and external IP 142.250.1.139. • Why: To validate the effectiveness of custom IDS rules in detecting outbound HTTP requests.
Additional notes	Utilized jq to isolate Flow IDs (e.g., 743523603871893) and confirmed the destination port was 80. Verified that the rule logic correctly identified the traffic flow.

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Task 3. Examine eve.json output

In this task, you must examine the additional output that Suricata generates in the `eve.json` file.

As previously mentioned, this file is located in the `/var/log/suricata/` directory.

The `event.json` file is the standard and main Suricata log file and contains a lot more data than the `fast.log` file. This data is stored in a JSON format, which makes it much more useful for analysis and processing by other applications.

1. Use the `cat` command to display the entries in the `eve.json` file:

```
cat /var/log/suricata/eve.json
```

The output returns the raw content of the file. You'll notice that there is a lot of data returned that is not easy to understand in this format.

Figure 7: Utilization of the cat command to inspect fast.log, confirming the generation of "GET on wire" alerts (Signature ID: 12345) based on HTTP traffic

What is the destination IP address listed for the last event in the 'eve.json' file?

Figure 8: Execution of a refined jq query to extract flow-specific metadata from eve.json, identifying destination IP 142.250.1.139 and correlating the unique Flow ID for forensic analysis.

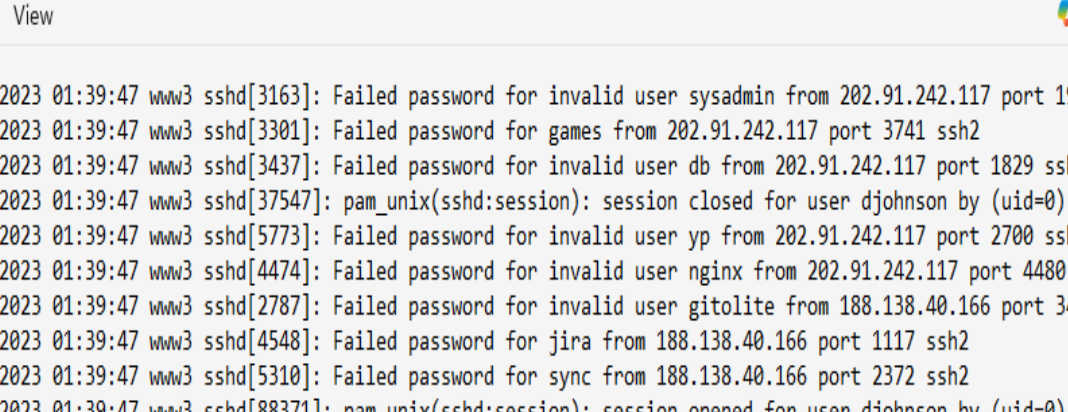
Lab 6: Security Monitoring and Querying with Wazuh SIEM

Scenario: I am acting as a Security Analyst for Buttercup Games. I have been tasked with investigating the **mailsv** server to identify any unauthorized access attempts or security issues within the provided tutorialdata.zip dataset.

Date: January 4, 2026..	Entry: #6
Description	Simulated Threat Hunt: Analyzed historical security telemetry from the Buttercup Games dataset using the Wazuh dashboard. Performed advanced queries to isolate unauthorized access attempts on the mail server.
Tool(s) used	Wazuh SIEM, Kibana Query Language (KQL)
The 5 W's	• Who: Potential malicious actors or automated scripts. • What: Identified over 300 failed SSH login attempts for the "root" account. • When: Historical activity captured within the provided logs. • Where: The www3 web server • Why: Brute-force attack pattern targeting administrative credentials.
Additional notes	Utilized complex Boolean logic (host.keyword: mailsv AND (fail* OR failed) AND root) to filter 100,000+ events down to 300+ critical alerts.

Lab Evidence

Figure 9: Manual analysis of web server logs (www3) showing repeated brute-force SSH attempts.



```
[Thu Feb 27 2023 01:39:47 www3 sshd[3163]: Failed password for invalid user sysadmin from 202.91.242.117 port 1994 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[3301]: Failed password for games from 202.91.242.117 port 3741 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[3437]: Failed password for invalid user db from 202.91.242.117 port 1829 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[37547]: pam_unix(sshd:session): session closed for user djohnson by (uid=0)
Thu Feb 27 2023 01:39:47 www3 sshd[5773]: Failed password for invalid user yp from 202.91.242.117 port 2700 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[4474]: Failed password for invalid user nginx from 202.91.242.117 port 4480 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[2787]: Failed password for invalid user gitolite from 188.138.40.166 port 3414 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[4548]: Failed password for jira from 188.138.40.166 port 1117 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[5310]: Failed password for sync from 188.138.40.166 port 2372 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[88371]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Feb 27 2023 01:39:47 www3 sshd[4113]: Failed password for root from 188.138.40.166 port 3457 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[3717]: Failed password for root from 188.138.40.166 port 3244 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[2822]: Failed password for invalid user postgres from 188.138.40.166 port 1697 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[3434]: Failed password for invalid user library from 188.138.40.166 port 3644 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[16848]: Received disconnect from 10.3.10.46 11: disconnected by user
Thu Feb 27 2023 01:39:47 www3 sshd[2847]: Failed password for invalid user varnish from 188.138.40.166 port 3717 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[4210]: Failed password for sync from 188.138.40.166 port 2101 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[1063]: Failed password for invalid user admin from 188.138.40.166 port 1834 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[3688]: Failed password for nagios from 188.138.40.166 port 1113 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[1073]: Failed password for invalid user alex from 209.160.24.63 port 3688 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[5251]: Failed password for invalid user sunny from 209.160.24.63 port 2713 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[2812]: Failed password for invalid user vmware from 209.160.24.63 port 2646 ssh2
Thu Feb 27 2023 01:39:47 www3 sshd[1686]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Feb 27 2023 01:39:47 www3 sshd[53851]: Failed password for invalid user jean-luc from 209.160.24.63 port 1224 ssh2
```

Reflections/Notes

1. On the Incident Response Lifecycle:

This project allowed me to apply the NIST Incident Response Lifecycle (Preparation, Detection & Analysis, Containment/Eradication/Recovery, and Post-Incident Activity) across multiple technical scenarios. Moving from the packet-level analysis of tcpdump to the broad visibility of Wazuh SIEM taught me that effective defense requires both granular investigation and high-level event correlation.

2. Core Skills Developed:

The most challenging but rewarding skill was learning to use Boolean logic and KQL queries to filter 100,000+ logs down to a specific security incident. Additionally, performing manual analysis on raw secure.log files confirmed that a Security Analyst must be able to interpret data in its simplest form to verify what automated tools are reporting.

3. Final Takeaway:

Completing this journal has reinforced the importance of the 'Five W's' (Who, What, When, Where, Why) in technical documentation. Standardizing how I report findings ensures that security investigations are clear, auditable, and actionable for the rest of the security team.